

The Geometric Trojan Horse: Vulnerabilities in Nested Spatial Protocol Geometries

Abstract

This paper analyzes a theoretical mathematical security model in which data validation is performed through geometric resonance within a nested spatial protocol, rather than traditional byte or hash-based checksums. We examine a specific theoretical construct known as a 12-nested 64-voxel cube protocol with a 72-bit addressable capacity, evaluating the mechanics of its mapping into $\mathcal{P}(\mathbb{R}^{24})$ and its Lattice Prime Sieve filter. Furthermore, we identify a critical vulnerability within this architecture—the "Geometric Trojan Horse"—which arises from a topological deficit where valid outer boundaries can encapsulate unverified, malicious inner geometries. Finally, we propose a formal recursive verification condition to mitigate this vulnerability.

1. Introduction

In traditional network security, data packets are validated as sequences of bits, bytes, or via cryptographic hashes. This paper examines an alternative, speculative security paradigm where data is treated as geometric forms. Under this architecture, a packet maps into a geometric coordinate space, its structure must satisfy specific resonance or lattice rules, and only "geometrically stable" packets are authorized for transmission.

This framework is defined by a mapping function Φ that projects standard encoded data into subsets of a 24-dimensional real coordinate space:

$$\Phi : \{0, 1\}^* \rightarrow \mathcal{P}(\mathbb{R}^{24})$$

Consequently, a data packet P transcends a mere sequence of symbols; it becomes a distinct geometric representation $G(P) = \Phi(P) \subseteq \mathbb{R}^{24}$. We define the allowed set of geometrically admissible packet states as a family of valid geometries:

$$\Lambda \subseteq \mathcal{P}(\mathbb{R}^{24})$$

2. System Architecture

2.1 The Lattice Prime Sieve

Network consensus in this system utilizes a lattice prime sieve over geometric subsets of $\mathbb{R}^{24}$. Let $\sigma : \mathcal{P}(\mathbb{R}^{24}) \rightarrow \mathbb{Z}$ be an integer-valued structural invariant extracted from the geometry. A packet P produces a structural value:

$$S(P) = \sigma(\Phi(P)) \in \mathbb{Z}$$

The system ensures that $S(P)$ avoids forbidden prime-modular collision zones. Let $\mathcal{P}_0 = \{p_1, p_2, \dots, p_k\}$ be the finite set of primes used by the sieve. For each prime p_i , we define a forbidden residue set $F_{p_i} \subseteq \mathbb{Z}/p_i\mathbb{Z}$.

A stable packet is defined as one that does not fall into these forbidden residue classes:

$$S(P) \bmod p_i \notin F_{p_i} \quad \text{for all } p_i \in \mathcal{P}_0$$

2.2 The Cosmological ID (Cosmo-ID)

The protocol utilizes a 64-bit identifier, known as the **Cosmo-ID**, to rapidly assert network consensus over the geometric states. However, as established later in Section 2.4, the full nested architecture possesses a 72-bit address space.

By the pigeonhole principle ($2^{72} > 2^{64}$), the 64-bit Cosmo-ID cannot serve as a unique enumeration of all possible nested states. Instead, it functions as a compressed consensus

label. Mathematically, it operates as a mapping from the full address space $\mathcal{A}_{12}$ to a reduced label space $\mathcal{T}_{64}$:

$$\text{Cosmo-ID} : \mathcal{A}_{12} \rightarrow \{0, 1\}^{64}$$

Because of this compression, an average of $2^8 = 256$ different full nested states map to the exact same Cosmo-ID. The network relies on this identifier for rapid sieve validation (ensuring the compressed structural value avoids forbidden primes). Consequently, the Cosmo-ID is computationally collision-prone by design. This distinction between the 64-bit identifier and the 72-bit true geometric state is a fundamental enabler of the Trojan vulnerability.

2.3 The 64-Voxel Cube Protocol

The fundamental data unit is a 64-voxel cube, defined as a $4 \times 4 \times 4$ voxel structure:

$$4^3 = 64$$

Let this discrete spatial container be formally denoted as:

$$C_{64} = \{v_{ijk} : 0 \leq i, j, k \leq 3\}$$

2.4 The 12-Nested System

The architecture is inherently recursive, utilizing 12 layers of nested containers. The nested address space is defined as the Cartesian product:

$$\mathcal{A}_{12} = C_{64}^{12}$$

Given that each layer has 64 possible voxel positions, the total number of possible nested addresses is:

$$|\mathcal{A}_{12}| = 64^{12} = (2^6)^{12} = 2^{72}$$

A particular nested packet instantiates geometries sequentially:

$$G_1 \supset G_2 \supset \dots \supset G_{12}$$

where each G_d is the geometry selected by the address component $a_d \in C_{64}$.

3. Vulnerability Analysis: The Geometric Trojan Horse

The fundamental flaw in this protocol arises from the disparity between outer and inner geometric verification. The vulnerability emerges when the network validates only the outer geometry (G_{out}) while neglecting the inner payload (G_{in}).

3.1 Boundary Aliasing and Topological Deficits

The system evaluates whether the outer container exists within the family of safe geometries:

$$G_{\text{out}} \in \Lambda$$

However, it fails to assert:

$$G_{\text{in}} \in \Lambda$$

This mathematically flawed implication ($G_{\text{out}} \in \Lambda \setminus \text{centernot} \implies G_{\text{in}} \in \Lambda$) creates a "topological deficit." The attack relies on an explicit boundary collision where an aliased geometry mimics a safe geometry:

$$\partial G_{\text{safe}} = \partial G_{\text{malicious}}, \quad G_{\text{safe}} \in \Lambda, \quad G_{\text{malicious}} \notin \Lambda$$

3.2 Hash Collisions and τ -Labels

The system utilizes a structural label, $\tau(G)$ (often embodied by the Cosmo-ID), to assert state validity. While an ideal system guarantees perfect injectivity ($\tau(G_1) = \tau(G_2) \implies G_1 = G_2$), this is mathematically impossible for a finite 64-bit label space applied to a 72-bit geometric space.

The Geometric Trojan Horse exploits this inherent lack of computational collision resistance, achieving a fixed-point state where a hidden structural perturbation δ does not alter the observed label:

$$\tau(G_{\text{safe}} + \delta) = \tau(G_{\text{safe}}) \quad \text{with} \quad \delta \neq 0$$

3.3 Recursive Verification Failure

For each layer d , we define the local validity condition $L_d(G_d)$:

$$L_d(G_d) = 1 \iff \forall p_i \in \mathcal{P}_0, \quad \sigma(G_d) \bmod p_i \notin F_{d,p_i}$$

A secure recursive verification function $V_d(G_d)$ requires a base case for the deepest layer:

$$V_{12}(G_{12}) = L_{12}(G_{12})$$

Followed by the recursive check for all preceding layers:

$$V_d(G_d) = L_d(G_d) \wedge V_{d+1}(G_{d+1})$$

In a compromised network, a flawed verifier stops early at some depth $m < 12$:

$$V_{\text{bad}}(P) = \bigwedge_{d=1}^m L_d(G_d)$$

4. Attack Execution and Local Failure

The exploitation phase proceeds mathematically as follows:

1. **Valid Outer Construction:** The attacker chooses an outer geometry G_1 such that it is globally valid ($L_1(G_1) = 1$).
2. **Malicious Nesting:** An inner payload P_{in} is embedded at some depth $m \in \{2, \dots, 12\}$ such that its geometry $G_m = \Phi(P_{\text{in}})$ violates the local condition: $L_m(G_m) = 0$.
3. **Signature Preservation:** The attacker ensures the root structural label remains unchanged despite the malicious nesting.
4. **Network Bypass:** The network checks only the outer layers $\bigwedge_{d=1}^{m-1} L_d(G_d) = 1$ and accepts the packet. Thus, $V_{\text{bad}}(P) = 1$.
5. **Local Execution:** A receiving node unwraps the nested payload P_{in} and evaluates its geometry $G_{\text{in}} = \Phi(P_{\text{in}})$.
6. **System Failure:** Because $G_{\text{in}} \notin \Lambda$ (specifically, $\sigma(G_{\text{in}}) \bmod p_i \in F_{p_i}$ for some prime p_i), the payload triggers invalid coordinate execution. In computing terms, this translates to parser confusion or local execution failure.

5. Proposed Mitigation and Security Conditions

To secure the 12-nested 64-voxel protocol, verification must be entirely recursive. The overarching security condition must satisfy:

$$V_{\text{secure}}(P) = \bigwedge_{d=1}^{12} L_d(G_d)$$

Correspondingly, structural τ -labels must be strictly chained from the deepest layer outward

using a collision-resistant function H :

$$\tau_{12} = H(G_{12})$$

and for $d = 11, 10, \dots, 1$:

$$\tau_d = H(G_d, \tau_{d+1})$$

This chaining ensures that any topological manipulation at layer G_{12} fundamentally changes the root label τ_1 . Specifically:

$$G_j \neq G'_j \implies \tau_1(G_1, \dots, G_j, \dots, G_{12}) \neq \tau_1(G_1, \dots, G'_j, \dots, G_{12})$$

This causes the modified packet to fail verification unless the adversary can find a computationally infeasible collision in H .

6. Conclusion

The "Geometric Trojan Horse" illustrates a sophisticated class of hidden-payload attacks within nested spatial protocols. By exploiting incomplete recursion and boundary aliasing, an attacker can bypass global prime-modular sieves applied to invariants of geometries in $\mathcal{P}(\mathbb{R}^{24})$. Resolving this vulnerability requires abandoning boundary-only verification in favor of rigorous, fully chained recursive validation across all nested layers.